Discovering Stable PDEs from Data

Why rollout stability matters more than one-step accuracy

An Pham Sinjini Ayomide Abhishek

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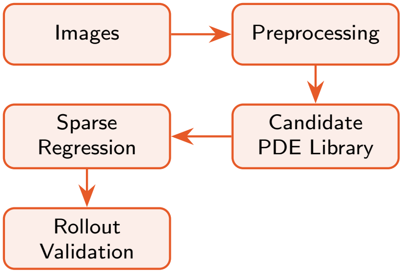
Problem & approach: stability-first PDE discovery

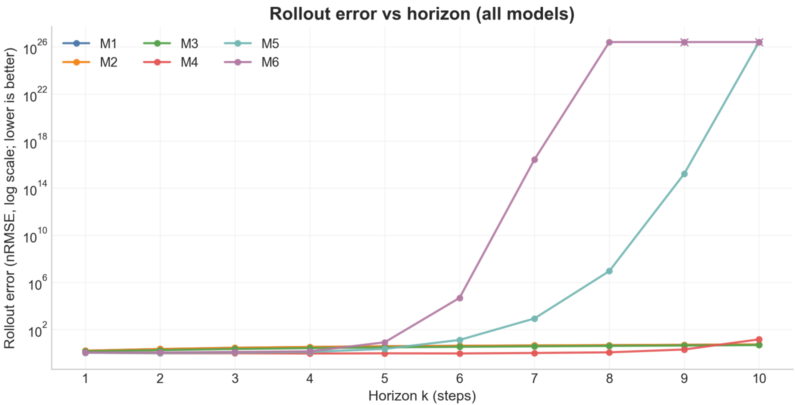
- Images are noisy $\rightarrow$ differentiation amplifies noise.
- Sparse regression identifies candidate PDE terms (e.g., STRidge).
- One-step accuracy can be misleading.
- **Rollout stability** is the real selection metric.

Goal: pick PDEs that simulate forward without blow-up.



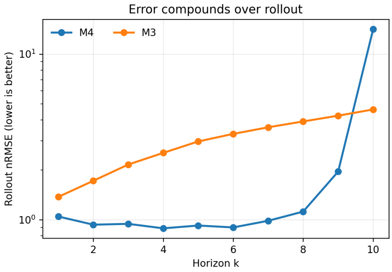
Rollout error vs horizon (all models)

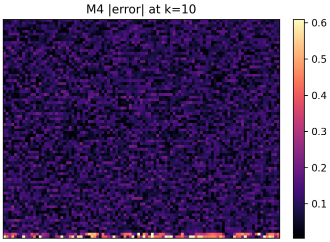
- Curves separate **stable** vs **blow-up** models.
- Focus on **growth rate** as horizon k increases.
- Blow-up $\Rightarrow$ unusable surrogate.



Insight: high one-step fit does *not* guarantee stable long-horizon rollouts.

Why one-step accuracy fails: errors compound





- Small errors compound over steps.
- Instability often starts locally.
- Then it spreads across the domain.

Message: multi-step rollout validation is required for usable PDE surrogates.

Best model: accuracy–stability trade-off

All models identified via STRidge on the same candidate library; selected by rollout.

Model variant (STRidge)	One-step R^2	Rollout nRMSE (k=10)
M4 (baseline, $+u^2$)	459.000×10^{-3}	14.100
M3 (+1st-order)	-1.180	4.630
M4 to_first ($\sigma = 2$) (chosen)	451.000×10^{-3}	7.480

Insight: model selection requires balancing accuracy and stability. High one-step R^2 does not guarantee stable rollouts—multi-step validation is essential.

Takeaways

Key message

- Separate **short-term fit** from **long-term stability**.
- Validate by multi-step rollouts, not just one-step metrics.
- **DMD consistently wins** on noisy/shifted data; ready for real images.

What can go wrong

A model can have strong one-step fit yet **diverge** during rollout (error compounding / blow-up).

What works better

STRidge-based identification + stabilization reduces long-horizon error; select by rollout first.

Context: High-res laser videos (2052×2560)

Noise: AWGN $\sigma \approx 8.5$; raw derivative variance 456.42.
extbfSolution: Anscombe + TV denoising.

Real images challenge

Even 100% synthetic error reduction may yield physically meaningless PDEs (unknown ground truth, complex noise, non-stationary dynamics).

Thank you for your attention!